

Ligjerata 2c

Shpejtësia këndore

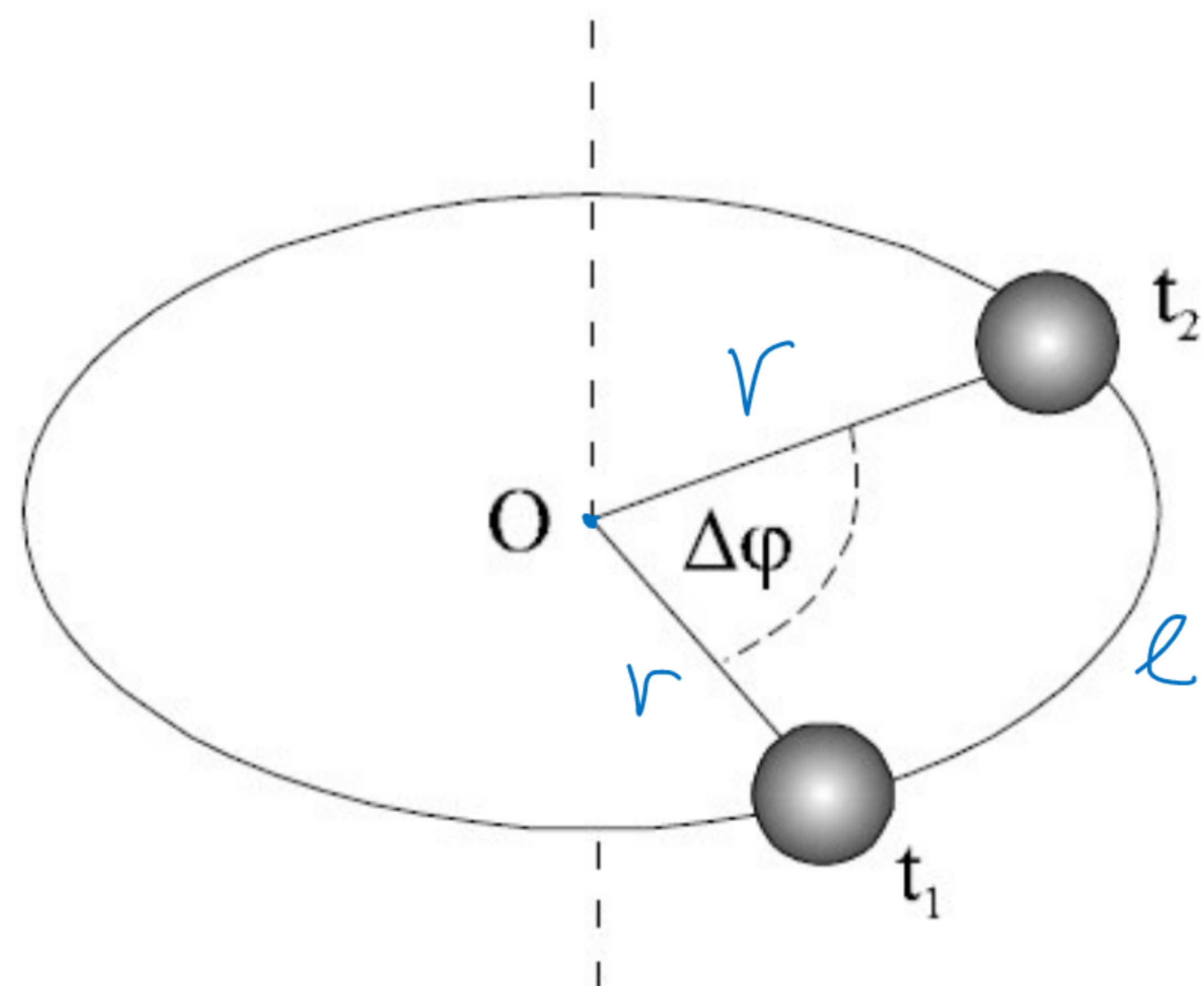
$$\Delta\varphi = \frac{l}{r} [\text{=}] \underline{\underline{\text{rad}}}$$

$$\omega_m = \langle \omega \rangle = \frac{\Delta\varphi}{\Delta t}$$

→ shpejtësia këndore

$$\Delta t \rightarrow 0$$

$$\lim_{\Delta t \rightarrow 0} \langle \omega \rangle = \lim_{\Delta t \rightarrow 0} \frac{\Delta\varphi}{\Delta t}$$



$$\omega = \frac{d\varphi}{dt} = \ddot{\varphi} \quad \text{shpejtësia këndore të çastit.}$$

$$\omega [\text{=}] \frac{d\varphi}{dt} [\text{=}] \frac{\text{rad}}{\text{s}} = \underline{\underline{\text{IS}}}$$

Nxitimi këndore

$$\alpha_m = \langle \alpha \rangle = \frac{\Delta\omega}{\Delta t}$$

$$\Delta t \rightarrow 0$$

$$\lim_{\Delta t \rightarrow 0} \langle \alpha \rangle = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} \quad \text{ose} \quad \alpha = \frac{d\omega}{dt} = \ddot{\omega} \quad \text{nxitimi i çastit.}$$

$$\text{nga: } \omega = \frac{d\varphi}{dt}$$

$$\alpha = \frac{d}{dt} \left(\frac{d\varphi}{dt} \right) = \frac{d^2\varphi}{dt^2} = \ddot{\varphi}$$

$$\alpha [\text{=}] \frac{d\omega}{dt} [\text{=}] \frac{\frac{\text{rad}}{\text{s}}}{\text{s}} = \underline{\underline{\frac{\text{rad}}{\text{s}^2}}} \quad \text{IS}$$